

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2024/2025

DECEMBER EXAMINATION

UCCD1213/UCCG1213 FUNDAMENTALS OF DIGITAL MEDIA TECHNOLOGY

FRIDAY, 27 DECEMBER 2024

TIME : 9.00 AM – 11.00 AM (2 HOURS)

EXAMINATION FOR BACHELOR DEGREE

Instruction to Candidates :

This question paper consists of **THREE (3)** questions in **Section A** and **TWO (2)** questions in **Section B**.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** question in **Section B**, marks will only be awarded for the **FIRST** question in that section in the order the candidate submits the answers.

Candidates are not allowed to use calculator.

Answer questions only in the answer booklet provided.

UCCD1213/UCCG1213 FUNDAMENTALS OF DIGITAL MEDIA TECHNOLOGY**Section A. Answer ALL Questions. [Total: 75 marks]**

Q1. (a) Explain **TWO (2)** benefits of using multimedia technology in the following industries: -

(i) Education. (4 marks)

(ii) Medical. (4 marks)

(b) List **FIVE (5)** considerations for designing the layout interface as a graphic designer. (5 marks)

(c) A storyboard is a graphic organizer of illustrations or images displayed sequentially to pre-visualize a motion picture, animation, or interactive media sequence. Discuss the following types of storyboards using appropriate examples: -

(i) Working storyboard. (4 marks)

(ii) Presentation storyboard. (4 marks)

(iii) Thumbnail or rough storyboard. (4 marks)

[Total: 25 marks]

Q2. (a) In your opinion, briefly explain **THREE (3)** reasons why non-technical users prefer to use an authoring system instead of programming to create digital content. (6 marks)

(b) A multimedia project consists of tasks combining various media elements, integrated through computer technology. It requires diverse skills from team members with varied backgrounds, and as a result, the multimedia environment embraces flexible approaches rather than a single, standardized workflow.

(i) Assess the importance of **FIVE (5)** key success factors in a multimedia project. (5 marks)

(ii) Which factor do you consider the most critical, and why? Support your evaluation with examples. (5 marks)

(c) Evaluate **THREE (3)** impacts of incorporating interactivity into a multimedia application in enhancing user engagement. (9 marks)

[Total: 25 marks]

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- Q3. (a) Differentiate between linear and non-linear text in multimedia applications.
(4 marks)
- (b) Break down **THREE (3)** steps in creating 3 Dimensional (3D) animation.
(9 marks)
- (c) Analyze **FOUR (4)** key factors to consider when incorporating video into a website. How might each factor affect user experience and website performance?
(8 marks)
- (d) List **FOUR (4)** types of transformations used to manipulate vector objects in graphic design.
(4 marks)
- [Total: 25 marks]

UCCD1213/UCCG1213 FUNDAMENTALS OF DIGITAL MEDIA TECHNOLOGY**Section B. Answer only ONE (1) Question. [Total: 25 marks]**

- Q4. (a) Discuss **FOUR (4)** issues and future directions related to the advancement of multimedia applications in today's digital world. (8 marks)
- (b) List **FOUR (4)** important parameters involved in calculating the size of a sound file. (4 marks)
- (c) Analyze **FOUR (4)** principle issues in designing website interfaces that may impact user experience and interface effectiveness. (8 marks)
- (d) Virtual reality (VR) is an immersive technology that simulates a realistic, computer-generated environment or a completely imaginary world, allowing users to interact with it as though they were physically present. Suggest **FIVE (5)** industries that can benefit significantly from VR technology. (5 marks)
[Total: 25 marks]
- Q5. (a) Distinguish **FOUR (4)** differences between raster and vector graphics. (8 marks)
- (b) List **FOUR (4)** types of multimedia authoring tools. (4 marks)
- (c) Analyze **ANY FOUR (4)** advantages of using digital video in education. (8 marks)
- (d) Augmented Reality (AR) is a technology that overlays digital content such as images, sounds, or other sensory enhancements onto the real world in real-time. AR can be experienced using devices like smartphones, tablets, smart glasses, or AR headsets. Provide **ANY FIVE (5)** examples of AR technologies currently used by people nowadays. (5 marks)
[Total: 25 marks]
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